

Auteur: Seda Jusjajeva
 Studierichting: Informatica
 Oefeningenreeks 4A nr. 38.8

$$\begin{aligned}
 & \int \frac{9x^3 - 21x^2 + 17x}{9x^2 - 12x + 5} dx \\
 & \int \frac{A(x)}{B(x)} dx = \int Q(x) dx + \int \frac{R(x)}{B(x)} dx \\
 & \begin{array}{r|l}
 \begin{array}{rrr}
 9x^3 & -21x^2 & +17x \\
 9x^3 & -12x^2 & +5x \\
 \hline
 & -9x^2 & +12x \\
 & -9x^2 & +12x & -5 \\
 \hline
 & & & 5
 \end{array}
 & \begin{array}{rrr}
 9x^2 & -12x & +5 \\
 & x & -1
 \end{array}
 \end{array} \\
 & \Rightarrow \int (x - 1) dx + \int \frac{5}{9x^2 - 12x + 5} dx \\
 & = \int x dx - \int dx + 5 \int \frac{dx}{9x^2 - 12x + 5} dx \\
 & = \frac{x^2}{2} - x + 5 \int \frac{dx}{9x^2 - 12x + 4 + 1} dx \\
 & = \frac{x^2}{2} - x + 5 \int \frac{dx}{(3x - 2)^2 + 1} dx
 \end{aligned}$$

Substitutie: $t = 3x - 2 \Rightarrow dt = 3dx$

$$\begin{aligned}
 & = \frac{x^2}{2} - x + 5 \int \frac{1}{t^2 + 1} \cdot \frac{1}{3} dt \\
 & = \frac{x^2}{2} - x + \frac{5}{3} \text{Bgtan } t + c \\
 & = \frac{x^2}{2} - x + \frac{5}{3} \text{Bgtan}(3x - 2) + c
 \end{aligned}$$

References